

Training Course

Amazon Web Service



- Deploying, administering, and monitoring AWS system and network resources in an automatable and reusable manner
- Prerequisites
 - ✓ Some Linux or Windows system administration experience
 - ✓ Basic familiarity with Linux command line or Windows PowerShell
 - ✓ Knowledge of popular development and scripting languages (e.g., Python) a plus, but not required

- **Module 1: System Operations on AWS**
 - ✓ **Goal:** Get started with the tools you need to administer resources on AWS
 - ✓ Data Center vs. the Cloud; High Level Overview; Securing Your AWS Account; Command Line Basics
- **Module 2: Computing on AWS**
 - ✓ **Goal:** Understand how to deploy instances and maintain instance health
 - ✓ Creating Instances, Instance Security, Pricing, Troubleshooting
 - ✓ **Lab 1:** Launching and Troubleshooting Amazon EC2 Instances

- Module 3: Networking on AWS
 - ✓ **Goal:** Understand how to create your own custom virtual private cloud using the AWS Management Console
 - ✓ Common Amazon VPC Scenarios, Example Walkthrough, Amazon VPC Peering and Direct Connect, Amazon VPC Security, Troubleshooting
 - ✓ **Lab 2:** Creating a New VPC, Verifying the VPC Configuration

- Module 4: Storage and Archiving in the Cloud
 - ✓ **Goal:** Use the various storage options in AWS to enable both short-term access and long-term storage of business data
 - ✓ Using Amazon Elastic Block Store, Amazon S3, AWS Import/Export, AWS Storage Gateway, Amazon Glacier, Storage Security, Troubleshooting, Storage Pricing Comparison
 - ✓ **Lab 3:** Using Storage in the Cloud, Synchronizing Files with Amazon S3

- Module 5: Monitoring in the Cloud
 - ✓ **Goal:** Perform basic monitoring with Amazon CloudWatch and Amazon CloudWatch Logs, and understand role of third party monitoring packages
 - ✓ Loggin Basics, Using CloudWatch and CloudWatch Logs, Using AWS CloudTrail, Security and Monitoring, Troubleshooting
 - ✓ **Lab 4:** Monitor EC2 System Status with CloudWatch; Auto-Stop Unused Instances

- Module 6: Managing Resource Consumption in the Cloud
 - ✓ **Goal:** Learn how to monitor resource spend and how to control costs via scripting and automation
 - ✓ Automating Cost Reduction, Tagging, CloudWatch, Cost Explorer, Trusted Advisor
 - ✓ **Lab 5:** Managing Resources with Tagging; Identifying and Terminating Non-Compliant Resources

- Module 7: Configuration Management in the Cloud
 - ✓ **Goal:** Cover the basics of using AMIs and Configuration Management to initialize new Amazon EC2 instances
 - ✓ Using and Creating New AMIs, Using Configuration Management, Troubleshooting CM

- Module 8: Creating Scalable Deployments in the Cloud
 - ✓ **Goal:** Use Elastic Load Balancing and Auto Scaling to scale systems out and in automatically
 - ✓ Common Deployment Scenarios, Redirecting Traffic with Elastic Load Balancing, Scaling to Demand with Auto Scaling, Routing with Amazon Route 53, Troubleshooting Scalable Deployments
 - ✓ **Lab 6:** Deploy an Auto Scaling Web Service

Course Schedule

Day	Presentations	Lab
Day 1	System Operations on AWS	
Day 2	Computing on AWS	X
Day 3	Networking on AWS	X
Day 4	Storage and Archiving in the Cloud	X
Day 5	Monitoring in the Cloud	X
Day 6	Managing Resource Consumption in the Cloud	X

Course Schedule

Day	Presentations	Lab
Day 7	Configuration Management in the Cloud Creating Scalable Deployment in the Cloud	X
Day 8, 9 & half of day 10	Assignment: + Introduce the assignment (purpose, requirements, ...) + Let student practice the assignment + Assessment, check and test practice + Summary content of course	X
Half of day 10	Test: multi choice quiz	

Course Schedule

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Day 4	Storage and Archiving in the Cloud	X
Day 5	Monitoring in the Cloud	X
Day 6	Managing Resource Consumption in the Cloud	X

Module 1:

Understanding System Operations on AWS



- **Goal:** Get started with the tools you need to administer resources on AWS
 - ✓ AWS Training and Certification
 - ✓ Comparing Data Center and the Cloud
 - ✓ Securing Your AWS Account
 - ✓ Command Line Basics

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AWS certification process

Available AWS Certifications

Professional

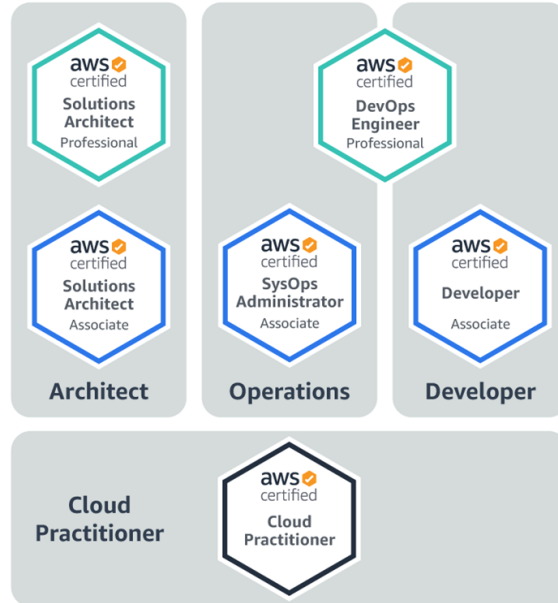
Two years of comprehensive experience designing, operating, and troubleshooting solutions using the AWS Cloud

Associate

One year of experience solving problems and implementing solutions using the AWS Cloud

Foundational

Six months of fundamental AWS Cloud and industry knowledge



aws certified
Updated May 2019

Specialty

Technical AWS Cloud experience in the Specialty domain as specified in the [exam guide](#)



AWS Document and References

<https://docs.aws.amazon.com/index.html>

<https://viblo.asia/newest>

<https://cafedev.vn/>

<https://udemy.com>

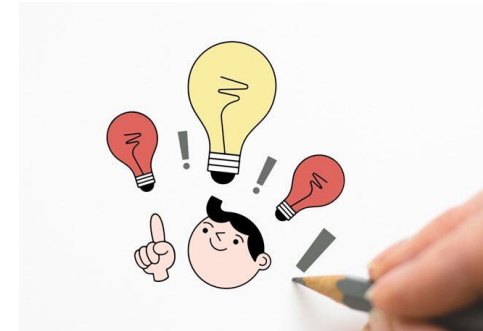
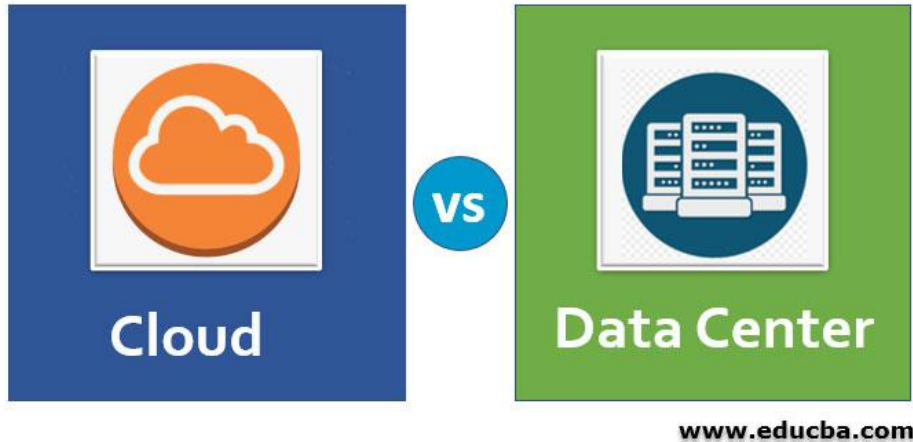
<https://acloudguru.com>

- **Goal:** Get started with the tools you need to administer resources on AWS
 - ✓ AWS Training and Certification
 - ✓ **Comparing Data Center and the Cloud**
 - ✓ High Level Overview
 - ✓ Securing Your AWS Account
 - ✓ Command Line Basics

What is System Operations?

- Configuration, deployment, and maintenance of system infrastructure
 - Network configuration and management
 - Server configuration and deployment
 - Application deployment and management
 - Storage, backup, and archive
 - Monitoring
 - Security

Cloud or Data Center?



System Operations in Data Center and The Cloud

Data Center

- Upfront capital expense
- Provision hardware and staff for normal operations and disaster recovery (DR)
- Limited experimentation and reusability

The Cloud

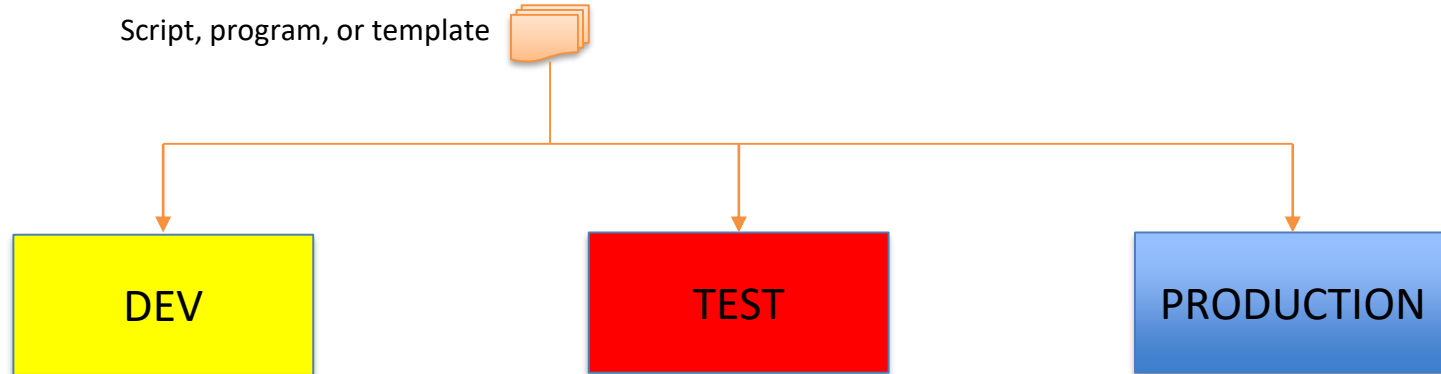
- Available when needed
- Build up, tear down and reuse with ease
- Reduced cost and planning for DR, storage redundancy
- More independence innovation within the company

Hybrid Model

- Connect data center and cloud resources

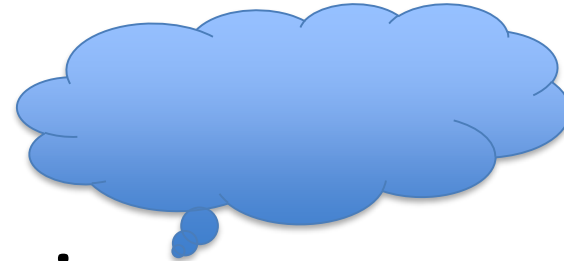
System Operations in the Cloud

- Cloud computing enables automated and repeatable deployment of infrastructure on demand.
- Systems can become self-describing.



System Operations Scenarios in the Cloud

Operation Area	Tasks
Create	Configure instances, store data, create reusable infrastructure, domains and routing
Deploy	Green field deployment, create dev/test environments, replicate across regions
Monitor	Log and monitor, capture log data, analyze, troubleshoot, scale to demand
Change	Deploy new version (e.g., “blue/green” deployment), maintain and patch instances reclaim resources



AWS Overview

What is AWS?

Cloud computing with AWS

Amazon Web Services (AWS) is the world's most comprehensive and broadly adopted cloud platform, offering over 200 fully featured services from data centers globally. Millions of customers—including the fastest-growing startups, largest enterprises, and leading government agencies—are using AWS to lower costs, become more agile, and innovate faster.

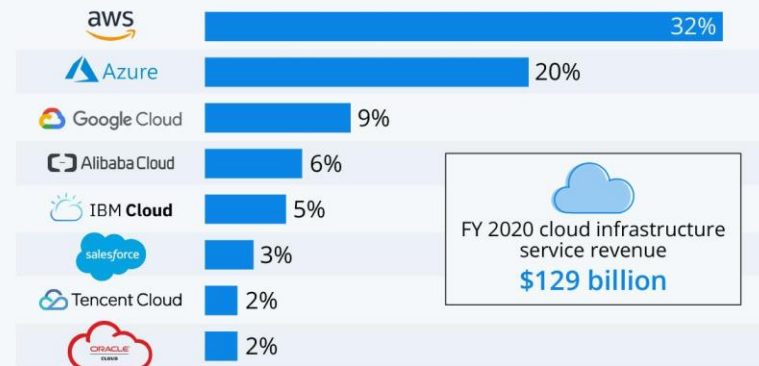
[Create an AWS Account](#)



AWS Market Share

Amazon Leads \$130-Billion Cloud Market

Worldwide market share of leading cloud infrastructure service providers in Q4 2020*



* includes platform as a service (PaaS) and infrastructure as a service (IaaS) as well as hosted private cloud services

Source: Synergy Research Group



statista

Global Infrastructure: AWS Regions, AZs, and Edge Locations

AWS achieves this by supporting a secure, redundant, and global infrastructure, which divide into regions, Availability Zones, and edge locations.

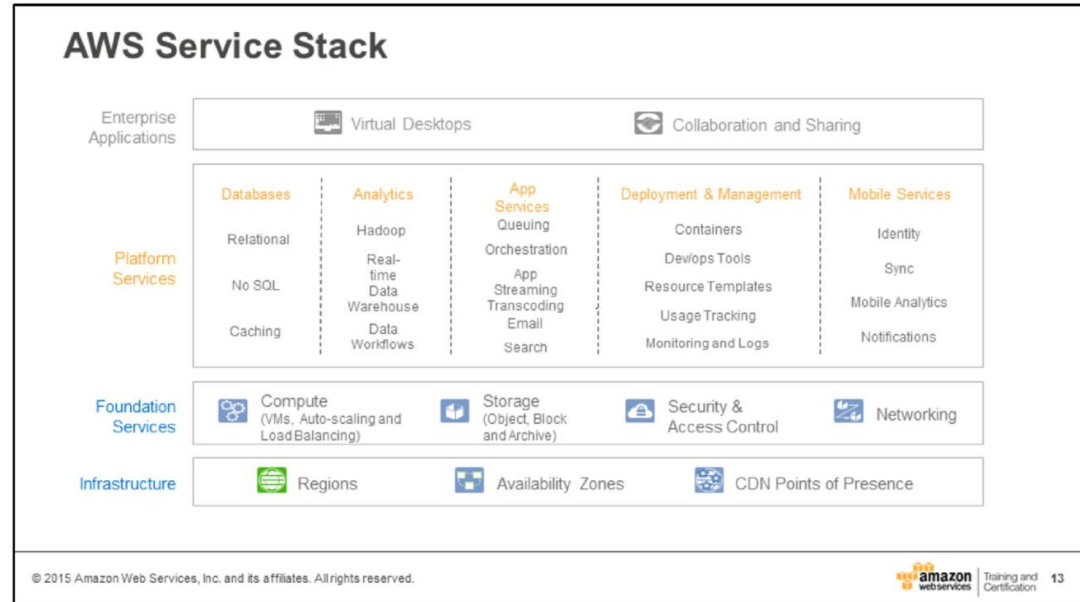
- A **region** is a collection of two or more Availability Zones in a specific geographic area.
- An **Availability Zone** is an isolated collection of AWS resources. Availability Zones within a region are connected through low-latency links.
- An **edge location** is A site that CloudFront uses to [cache](#) copies of your content for faster delivery to users at any location.

https://aws.amazon.com/about-aws/global-infrastructure/regions_az/?nc1=h_ls

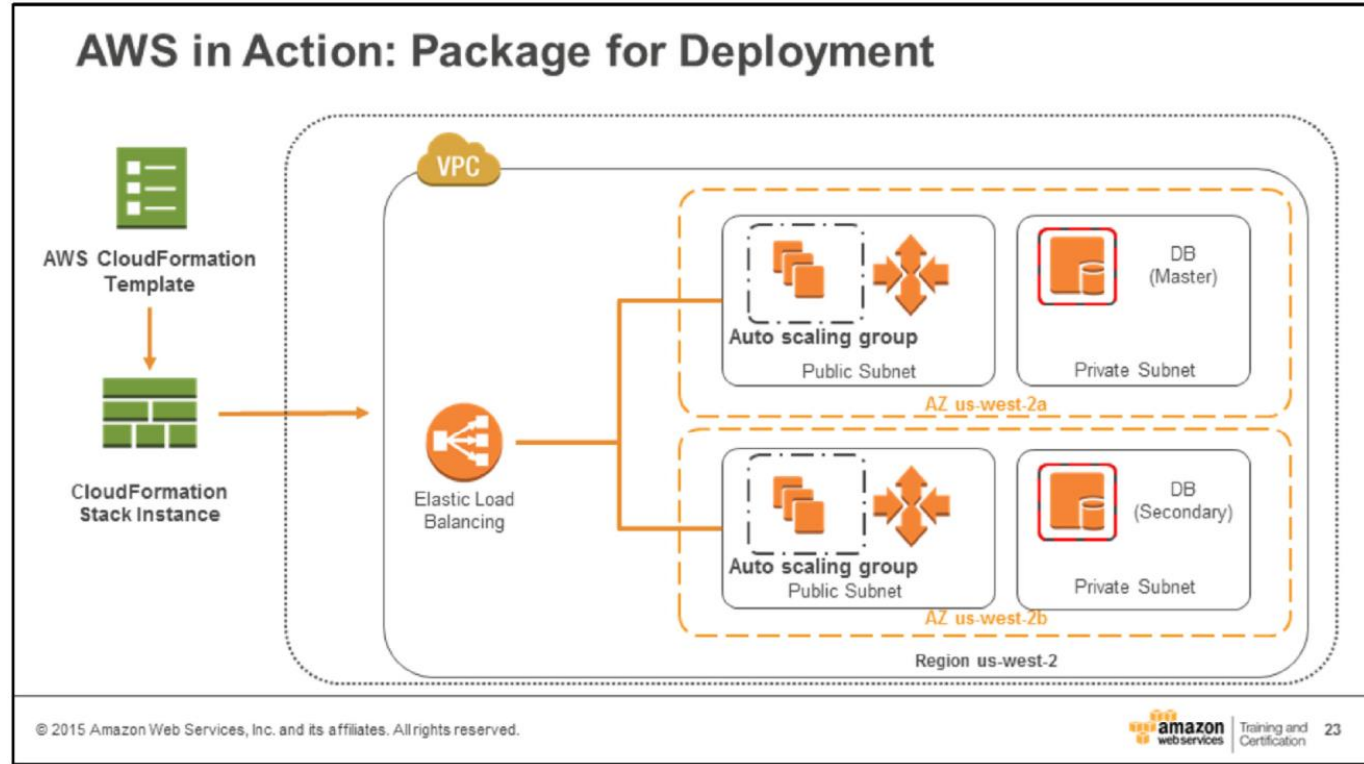
AWS Service Stack

AWS provides a highly reliable, scalable, low-cost infrastructure platform in the cloud that powers hundreds of thousands of businesses all over the world.

At the core is the compute, storage and data services that are the heart of AWS's offering.



AWS in Action Example



AWS in Action Example

Other Amazon Web Services Related to Systems Operations



Amazon Relational Database Service (RDS)

A web service that makes it easy to set up, operate, and scale a relational database in the cloud



Amazon Simple Notification Service (SNS)

Feed notifications of critical system events to internet connected smart devices, monitored aliases, third party monitoring systems



AWS Identity and (IAM) Access Management

Securely control access to Amazon Web Services and resources for your users



Amazon Simple Queue Service (SQS)

Post system events to a job queue for asynchronous processing

...and many others

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When create AWS resources, those resources exist at a scope which varies depending on the service – either global, region, or Availability Zone. Resources with global and regional scope are automatically distributed across multiple Availability Zones by AWS.

Scope of Amazon Web Services

Scope	AWS Resource
Global	• IAM Users, Groups, and Roles • Amazon Route 53 Hosted Zones and Record Sets • CloudFront Distributions
Region	• Amazon S3 Buckets • Amazon Machine Images (AMIs) • CloudWatch Metrics • Amazon EBS Snapshots • Amazon ElastiCache Clusters • Virtual Private Cloud (VPC)
Availability Zone	• Amazon EC2 Instances • Amazon EBS Volumes • Amazon RDS Database Instances • Subnets

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- **Goal:** Get started with the tools you need to administer resources on AWS
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 - ✓ High Level Overview
 - ✓ **Securing Your AWS Account**
 - ✓ Command Line Basics

Identity and Access Management (IAM) Overview

IAM features

- Centralized access control for AWS
- Integrated with AWS
- Fine-grained access control
- Secure by default (restricted unless explicitly granted access)
- Free as a service

What you can do with IAM:

- Define group permissions by service and resource.
- Create and manage users and assign to groups.
- Enable federation to your corporate directory.
- Create IAM roles for Amazon EC2 instances and other uses.

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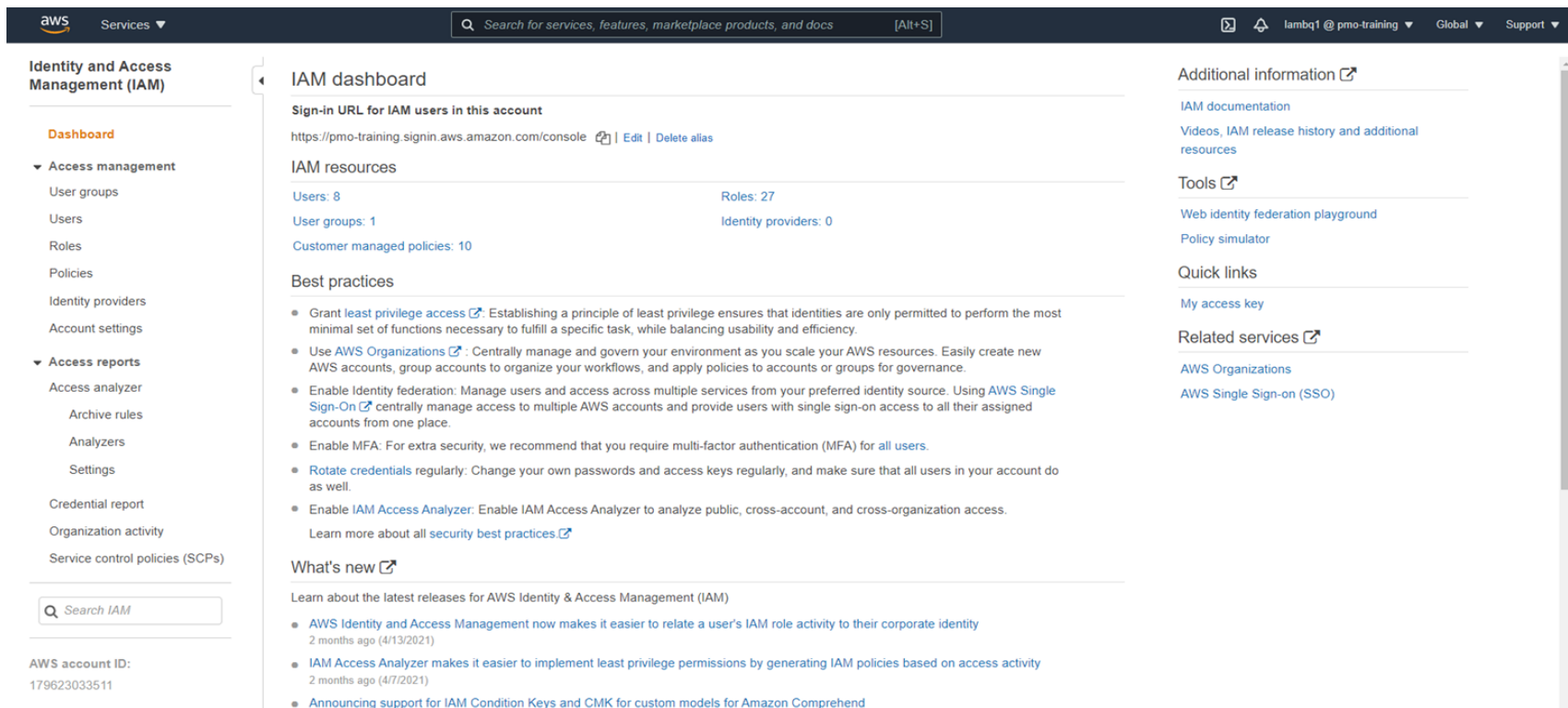


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AWS Identity and Access Management (IAM) provides a centralized environment in which to administer users, groups, roles and permissions.

Module 1: Understand System Operations in AWS

Access to IAM services > Login AWS account >> Service > IAM (Manage access to AWS resources)



The screenshot shows the AWS IAM dashboard. The left sidebar contains the 'Identity and Access Management (IAM)' menu with options like Dashboard, Access management, Access reports, and Account settings. The main content area displays the 'IAM dashboard' with a sign-in URL, IAM resources summary (8 users, 27 roles, 1 user group, 0 identity providers, 10 policies), best practices, and a 'What's new' section. The right sidebar provides additional information, tools, quick links, and related services. The top navigation bar includes the AWS logo, search bar, and user profile.

Identity and Access Management (IAM)

- Dashboard
- Access management
 - User groups
 - Users
 - Roles
 - Policies
 - Identity providers
 - Account settings
- Access reports
 - Access analyzer
 - Archive rules
 - Analizers
 - Settings
 - Credential report
 - Organization activity
 - Service control policies (SCPs)

Search IAM

AWS account ID: 179623033511

IAM dashboard

Sign-in URL for IAM users in this account

<https://pmo-training.signin.aws.amazon.com/console> [Edit](#) [Delete alias](#)

IAM resources

Users: 8	Roles: 27
User groups: 1	Identity providers: 0
Customer managed policies: 10	

Best practices

- Grant [least privilege access](#): Establishing a principle of least privilege ensures that identities are only permitted to perform the most minimal set of functions necessary to fulfill a specific task, while balancing usability and efficiency.
- Use [AWS Organizations](#): Centrally manage and govern your environment as you scale your AWS resources. Easily create new AWS accounts, group accounts to organize your workflows, and apply policies to accounts or groups for governance.
- Enable [IAM Single Sign-On](#): Manage users and access across multiple services from your preferred identity source. Using [AWS Single Sign-On](#) centrally manage access to multiple AWS accounts and provide users with single sign-on access to all their assigned accounts from one place.
- Enable [MFA](#): For extra security, we recommend that you require multi-factor authentication (MFA) for [all users](#).
- [Rotate credentials](#) regularly: Change your own passwords and access keys regularly, and make sure that all users in your account do as well.
- Enable [IAM Access Analyzer](#): Enable IAM Access Analyzer to analyze public, cross-account, and cross-organization access.

Learn more about all [security best practices](#)

What's new

Learn about the latest releases for AWS Identity & Access Management (IAM)

- [AWS Identity and Access Management now makes it easier to relate a user's IAM role activity to their corporate identity](#)
2 months ago (4/13/2021)
- [IAM Access Analyzer makes it easier to implement least privilege permissions by generating IAM policies based on access activity](#)
2 months ago (4/7/2021)
- [Announcing support for IAM Condition Keys and CMK for custom models for Amazon Comprehend](#)

Additional information

- [IAM documentation](#)
- [Videos, IAM release history and additional resources](#)

Tools

- [Web identity federation playground](#)
- [Policy simulator](#)

Quick links

- [My access key](#)

Related services

- [AWS Organizations](#)
- [AWS Single Sign-on \(SSO\)](#)

IAM enables a customer to create multiple users and manage the permissions for each of these users within their AWS account.

IAM User Administration

- 📦 Add/delete users and assign to groups.
- 📦 Manage user passwords including strength, length, and password reset.
- 📦 Administer user access keys (create/revoke).
- 📦 Administer multi-factor authentication (MFA) for privileged users.
- 📦 Provide individualized permissions by service and resource.

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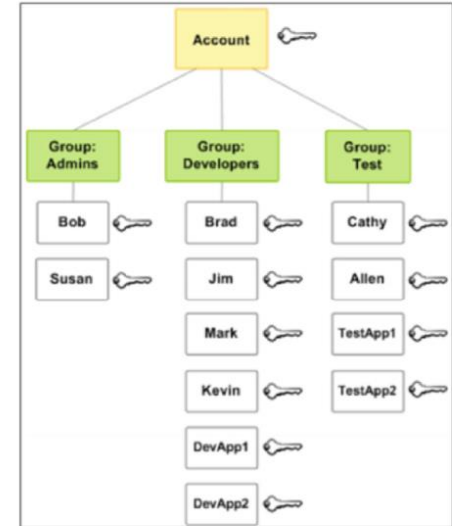


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IAM Groups

- ❏ Create IAM groups for functions, departments, and levels of responsibility.
- ❏ Attach template policies by service.
 - ReadOnlyAccess
 - FullAccess
- ❏ Allow and deny access by service and resource.
- ❏ Re-assign users to different groups as needed.



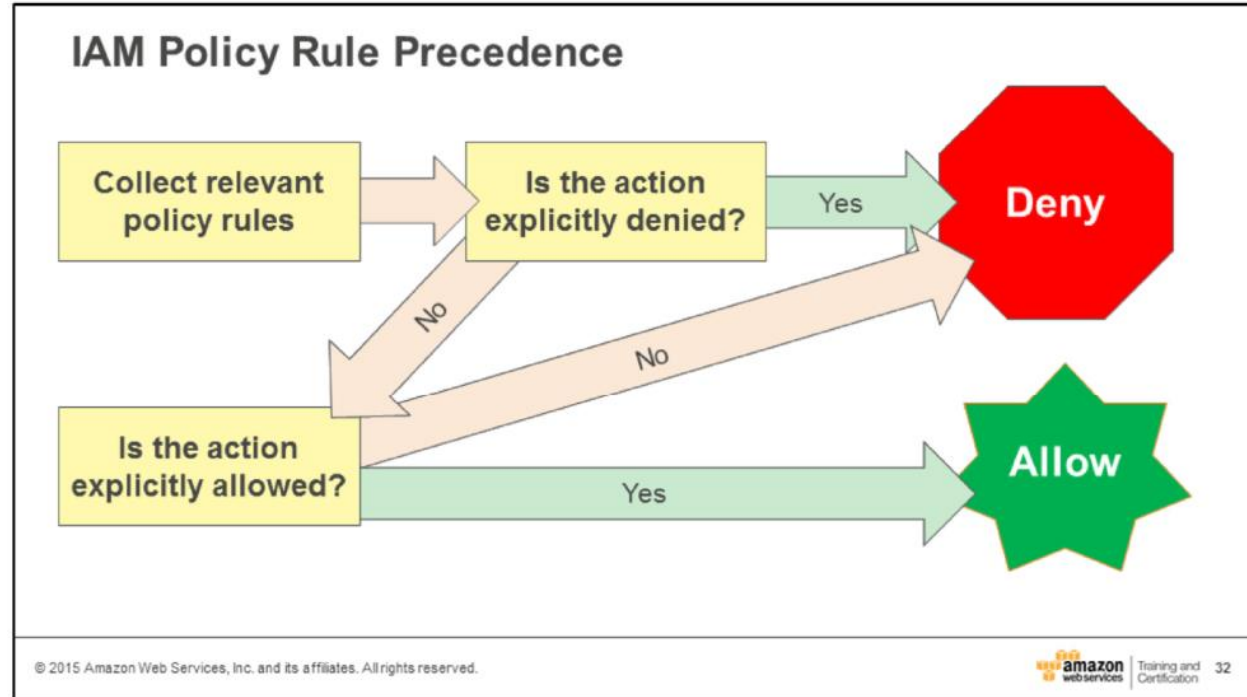
Using group simplifies the task of adding or removing permissions to a large number of users simultaneously.

IAM User Permissions

- ❏ By default, a new user can't do anything with any service.
- ❏ Access must be explicitly provided to users as individuals or as members of a group.
- ❏ Assign users to groups with pre-specified permissions.
- ❏ Use a “least privilege” policy when assigning permissions (just enough access to do their job).
 - The chance of making mistakes is reduced.
 - Security is easier to loosen than tighten.
 - Permissions are automatically triggered (If they need it, they will ask for it).

To determine whether the request should be allowed or denied, these rules are followed:

- By default, all requests are denied
- An explicit allow overrides this default
- An explicit deny overrides any allows



These days, password-only security is not enough. Should enable multi-factor authentication (MFA). With MFA, users must supply both a password and a one-time challenge/response token.

Multi-Factor Authentication (MFA)

Authentication

- Password (something you know)
- Token (something you have)

Types of MFAs

- Hardware (Gemalto)
- Virtual (e.g, Google Authenticator)

MFA device resync capability

Use when delegating access with IAM roles.



Securing your AWS Account

- ☐ Delegate system administration functions to least-privilege IAM admin groups
- ☐ Require multi-factor authentication for root-level access
- ☐ Physically secure hardware MFA devices in a secure place such as a vault
- ☐ Do not share root credentials with anyone other than the account holder
- ☐ Use IAM roles to provide cross-account access

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Using Amazon Web Services

Three Ways to Use AWS



AWS Management Console

Easy-to-use graphical interface that supports majority of Amazon Web Services.



Command Line Interface (CLI)

Access to services via discrete commands that can be issued from a Linux command line, Linux shell script, Windows cmd prompt, Windows batch file, or Windows Powershell.



Software Development Kits (SDKs)

Launch EC2 instances, configure networks, etc. from most major programming languages (Python, Ruby, .NET, Java, etc.).

Using Amazon Web Services

AWS CLI Basics

- ❖ Available for Linux, Windows, Mac.
- ❖ Can be installed and configured on Amazon EC2 instances automatically using user data (discussed later).
- ❖ Use the *aws configure* command to:
 - Setting access key and secret key, which is not necessary if IAM roles are used
 - Specify the default region name
 - Specify the default output format

Using Amazon Web Services

AWS CLI Basics

Command Line Format



Example: Amazon EC2

Service
(command)

Operation
(sub-command)

Parameters

Options

```
aws ec2 stop-instances --instance-id i-la2b3c4d --output json
```

```
aws ec2 run-instances --cli-input-json file://./lcf/webserver.json
```

```
$ aws help
```

```
$ aws ec2 help
```

```
$ aws ec2 describe-instances help
```

Using Amazon Web Services

AWS CLI Basics

Common Amazon EC2 and Amazon S3 Commands

Amazon EC2	Amazon S3 (High-level)
<code>aws ec2 run-instances</code>	<code>aws s3 ls</code>
<code>aws ec2 describe-instances</code>	<code>aws s3 cp</code>
<code>aws ec2 create-volume</code>	<code>aws s3 mv</code>
<code>aws ec2 create-vpc</code>	<code>aws s3 rm</code>

 Note that storage volumes (Amazon Elastic Block Store) and Amazon VPC are sub-commands of `aws ec2`

Using Amazon Web Services

AWS CLI Basics

Command Line Results

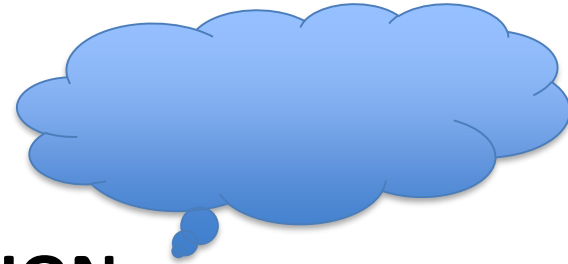
📦 Results by default returned in [Javascript Object Notation](#) (JSON) format

```
[ec2-user@ip-172-31-1-114 ~]$ aws ec2 describe-instances
{
  "Reservations": [
    {
      "OwnerId": "244698725403",
      "ReservationId": "r-31984d40",
      "Groups": [],
      "Instances": [
        {
          "Monitoring": {
            "State": "disabled"
          },
          "PublicDnsName": "ec2-54-165-109-159.compute-1.amazonaws.com",
          "State": {
            "Code": 16,
            "Name": "running"
          },
          "EbsOptimized": false,
          "LaunchTime": "2014-09-07T23:21:30.000Z",
          "PublicIpAddress": "54.165.109.159",
```

Requests configuration of all EC2 instances running in default region

Multiple items returned as arrays of values

Colon-separated name-value pairs



IAM SECTION

IAM: User & Groups

- IAM = Identity and Access Management, Global service
- Root account create by default, shouldn't be used or shared
- User are people within your organization, and can be grouped
- Groups only contain users, not other groups
- User don't have to belong to a group, and user can belong to multi group

IAM: User & Groups



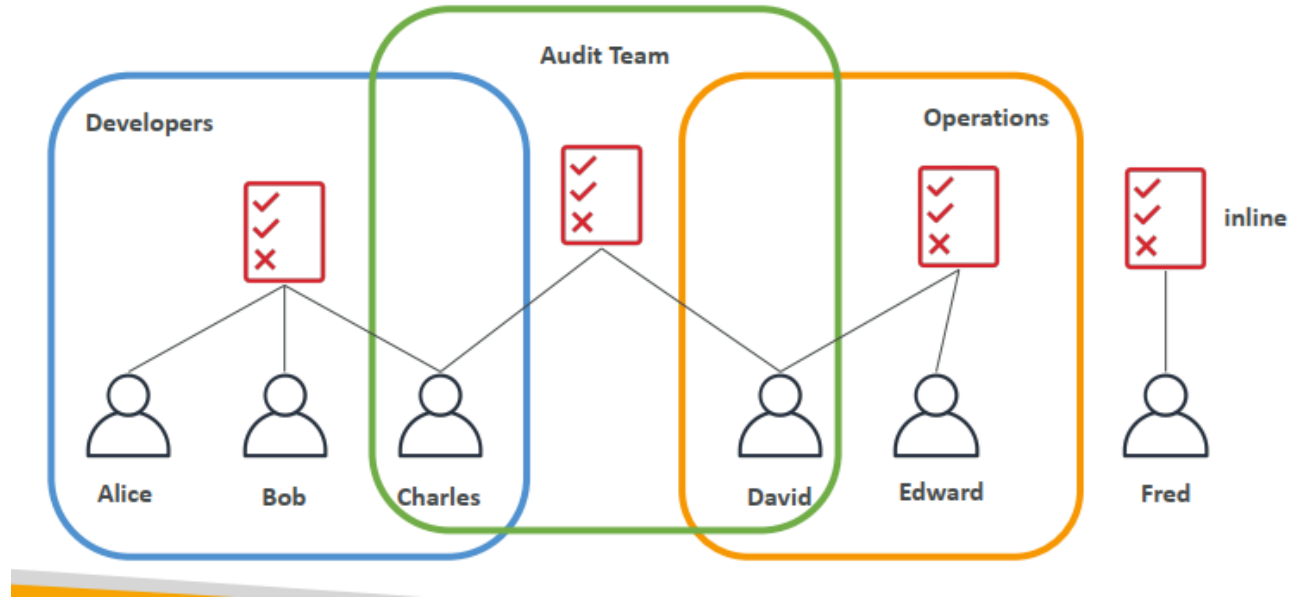
IAM: User & Groups

- Users or Groups can be assigned JSON documents called **policies**
- These **policies** define the permissions of the user
- In AWS you apply the **least privilege** principle: don't give more permission than user need

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Action": "ec2:Describe*",
      "Resource": "*"
    },
    {
      "Effect": "Allow",
      "Action": "elasticloadbalancing:Describe*",
      "Resource": "*"
    },
    {
      "Effect": "Allow",
      "Action": [
        "cloudwatch:ListMetrics",
        "cloudwatch:GetMetricStatistics",
        "cloudwatch:Describe*"
      ],
      "Resource": "*"
    }
  ]
}
```



IAM Policies inheritance



IAM Policies Structure

- Consist of
 - Version: policy language version, always include "2012-10-17"
 - Id: an identifier for the policy (optional)
 - Statement: one or more individual statements (required)
- Statement consist of
 - Sid: an identifier for the statement (optional)
 - Effect: whether the statement allows or denies access (allow, deny)
 - Principal: account/user/role which this policy applied to
 - Action: list of actions this policy allows or denies
 - Resource: list of resources to which the actions applied to
 - Condition: conditions for when this policy is in effect (optional)

```
{
  "Version": "2012-10-17",
  "Id": "S3-Account-Permissions",
  "Statement": [
    {
      "Sid": "1",
      "Effect": "Allow",
      "Principal": {
        "AWS": ["arn:aws:iam::123456789012:root"]
      },
      "Action": [
        "s3:GetObject",
        "s3:PutObject"
      ],
      "Resource": ["arn:aws:s3:::mybucket/*"]
    }
  ]
}
```

IAM – Password Policy

- Strong passwords = higher security for your account
- In AWS, you can setup a password policy
 - Set a minimum password length
 - Require specific character types
 - Including uppercase letters
 - Lowercase letters
 - Number
 - Non-alphanumeric characters
- Allow all IAM users to change their own passwords
- Require users to change their password after some time (password expiration)
- Prevent password re-use

Multi Factor Authentication - MFA

- Users have access to your account and can possibly change configurations or delete resources in your AWS account
- You want to protect your Root Accounts and IAM users
- MFA = password you know + security device you own
- Main benefit of MFA:
 - If a password is stolen or hacked, the account is not compromised



Alice

Password

+



=>

Successful login

MFA devices options in AWS

Virtual MFA device



Google Authenticator
(phone only)



Authy
(multi-device)

Support for multiple tokens on a single device.

Universal 2nd Factor (U2F) Security Key



YubiKey by Yubico (3rd party)

Support for multiple root and IAM users
using a single security key

MFA devices options in AWS

Hardware Key Fob MFA Device



Provided by Gemalto (3rd party)

Hardware Key Fob MFA Device for AWS GovCloud (US)



Provided by SurePassID (3rd party)

How can users access AWS?

- To access AWS, you have three options:
 - AWS Management Console (protected by password + MFA)
 - AWS Command Line Interface (CLI): protect by access keys
 - AWS Software Developer Kit (SDK) – for code: protected by access keys
- Access Keys are generated through the AWS Console
- Users manage their own access keys
- Access Keys are secret, just like a password. Don't share them
- Access Key ID ~= username
- Secret Access Key ~= password

Example (Fake) Access Keys

Access keys

Use access keys to make secure REST or HTTP Query protocol requests to AWS service APIs. For your protection, you should never share your secret keys with anyone. As a best practice, we recommend frequent key rotation. [Learn more](#)

Create access key

Access key ID	Created	Last used	Status	
AKIAASK4E37PV4TU3RD6C	2020-05-25 15:13 UTC+0100	N/A	Active	Make inactive ✕

- Access key ID: AKIAASK4E37PV4983d6C
- Secret Access Key: AZPN3z0jWozWCndljhB0Unh8239aIbzbzO5fqkZq
- Remember: don't share your access keys

What's the AWS CLI?

- A tool that enables you to interact with AWS services using commands in your command-line shell
- Direct access to the public APIs of AWS services
- You can develop scripts to manage your resources
- It's open-source <https://github.com/aws/aws-cli>
- Alternative to using AWS Management Console

What's the AWS SDK?

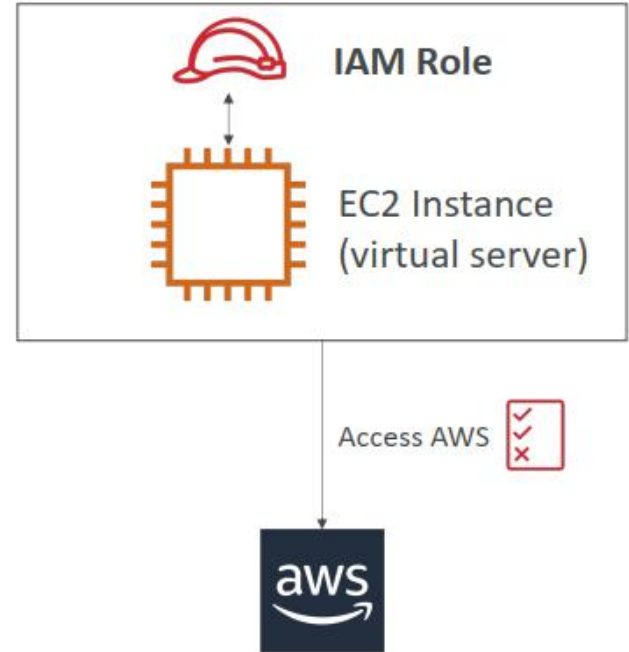
- AWS Software Development Kit (AWS SDK)
- Language-specific APIs (set of libraries)
- Enables you to access and manage AWS services programmatically
- Embedded within your application
- Supports
 - SDKs (JavaScript, Python, PHP, .NET, Ruby, Java, Go, Node.js, C++)
 - Mobile SDKs (Android, iOS, ...)
 - IoT Device SDKs (Embedded C, Arduino, ...)
- Example: AWS CLI is built on AWS SDK for Python



Your Application

IAM Roles for Services

- Some AWS service will need to perform actions on your behalf
- To do so, we will assign permissions to **AWS services** with **IAM Roles**
- Common roles:
 - **EC2 Instance Roles**
 - Lambda Function Roles
 - Roles for CloudFormation



IAM Security Tools

- **IAM Credentials Report (account-level)**
 - A report that lists all your account's users and the status of their various credential
- **IAM Access Advisor (user-level)**
 - Access advisor shows the service permission granted to a user and when those services were last accessed
 - You can use this information to revise your policies

IAM Guidelines & Best Practices

- Don't use the root account except for AWS account setup
- One physical user = One AWS **user**
- **Assign users to groups** and assign permissions to groups
- Create a **strong password policy**
- Use and enforce the use of **Multi Factor Authentication (MFA)**
- Create and use **Roles** for giving permissions to AWS services
- Use Access Keys for Programmatic Access (CLI/SDK)
- Audit permissions of your account with the IAM Credentials Report
- **Never share IAM users & Access Keys**

IAM Section - Summary

- **Users:** mapped to a physical user, has a password for AWS Console
- **Groups:** contains users only
- **Policies:** JSON document that outlines permissions for users or groups
- **Roles:** for **EC2 instances** or AWS services
- **Security:** MFA + Password Policy
- **Access Keys:** access AWS using the CLI or SDK
- **Audit:** IAM Credential Reports & IAM Access Advisor

Thank you

